MIL-D-37738
16 January 1978

MILITARY SPECIFICATION

DISH, CULTURE, PETRI, Top and Bottom,
Complete, Sterile, Disposable

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one size of sterile, plastic, disposable Petri culture dish that is suitable in performing bacteriological cultures and biological assays. The size of the Petri culture dish is commercially referred to as 150 x 15.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

MIL-S-36586 - Sterilization Test Strip Set, Bacterial Spore.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, CODE ATT, 2800 So. 20th St., Phila., PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6640

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards and publications required by contractors in connection with specific procurement function should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The Pharmacopeia of the United States of America

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042.)

3. REQUIREMENTS

3.1 Material. The Petri dish shall be fabricated from polystyrene that shall be clear and transparent, and free of toxic lubricants and other additives which would adversely affect the growth of bacterial cultures.

3.2 Design and construction. The design and dimensions of the Petri dish shall be in accordance with figure 1. Minor variations in design shall be accepted provided they do not conflict with dimensional or configurational limitations indicated in the figure. The bottom section shall be constructed so that it sits flat on a plane and level surface and provides for a uniform media depth. The top section shall fit over and shall be easily removed from the bottom section by using one hand.

3.2.1 Writing surface. The exterior surfaces of the Petri dish shall be such that markings applied with a wax or crayon type pencil are legible.

3.3 Sterility. The Petri dish shall be sterile when tested with a biological indicator as specified in 4.4.1.

3.3.1 Certificate of sterility. A certificate of sterility, stating that samples from each sterilizer lot have been tested and found to be sterile, shall be submitted to the procuring activity. The certificate of sterility shall include:

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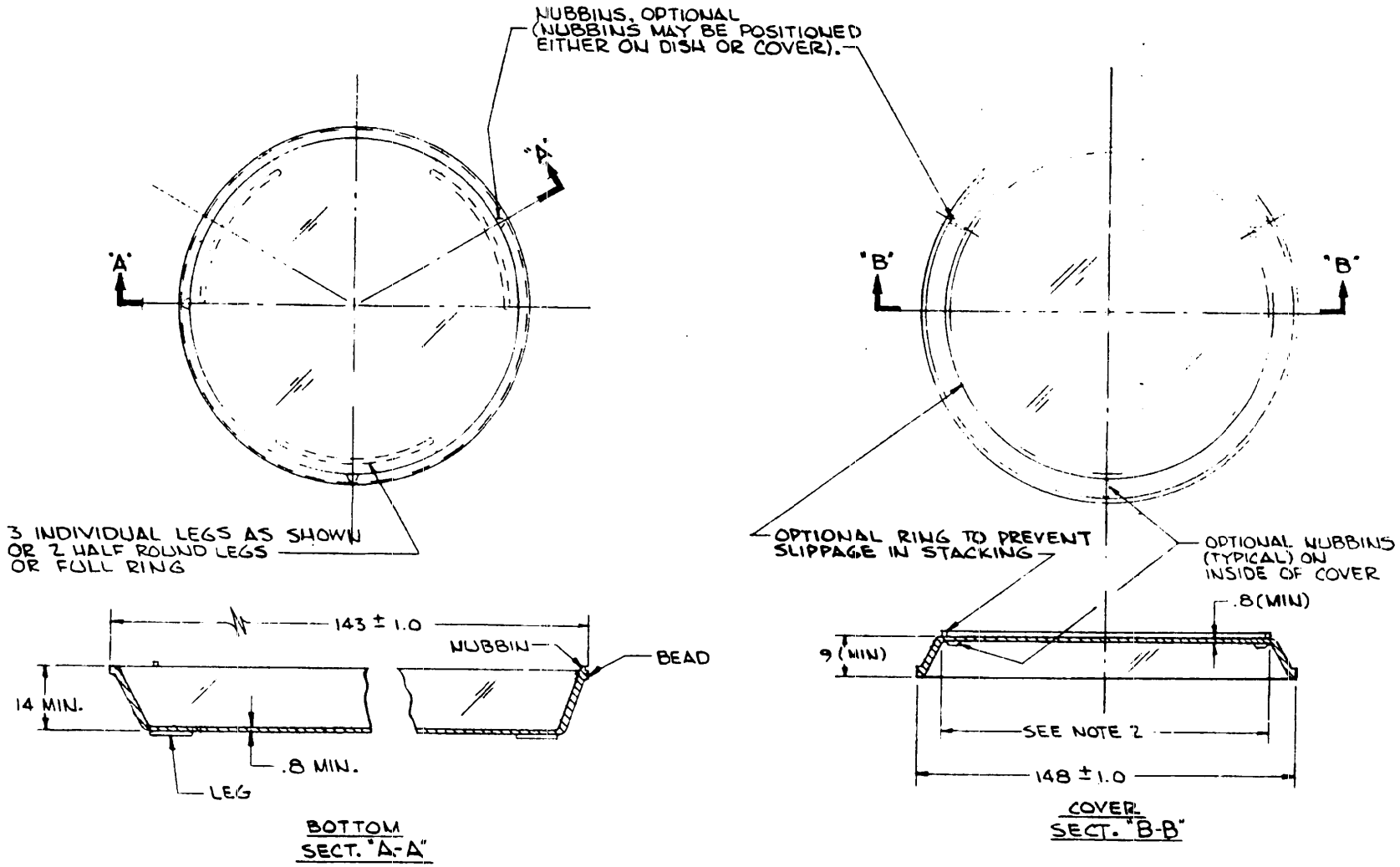
a. Item Identification.
b. National Stock Number (NSN).
c. Contract or purchase order number.
d. Sterility lot or control number.
e. Date of sterilization.
f. Quantity of product in sterilizer lot.
g. Type of sterilization (gas, radiation, heat, steam, etc.).
h. Form of biological indicator (i.e., inoculated paper spore strip or carrier, or inoculated product).
i. Test organism(s) and population density used.
j. Test medium, incubation temperature, incubation time, and dates of tests.
k. Number of test units.
l. Statement that samples comply with sterility tests.
m. Signature of head of laboratory or a designated qualified alternate.

When inspection is made at destination, the certificate of sterility shall be submitted by the contractor to the Directorate of Medical Materiel, ATTN: DPSC-ATQ, Defense Personnel Support Center, 2800 South 20th Street, Phila., Pa., 19101, and a copy forwarded with each shipment to the consignee. When inspection is made at source, certificate of sterility shall be furnished to the cognizant Government Quality Assurance Representative for submission to the procuring activity.

3.3.2 Sterility lot. A sterility lot is that single quantity of product subjected to the same manufacturing operation and simultaneously sterilized in the same sterilizer chamber. Each unit package shall be clearly marked with a sterility lot (control) number and date of sterilization as specified in 5.3.2. Each sterility lot shall be traceable to the pertinent manufacturing lot(s). In addition, each bag of dishes shall be clearly marked with a sterility lot (control) number as specified in 5.3.1.

3.4 Workmanship. Workmanship shall be first class throughout. The Petri dish shall be supplied clean and dry, and shall be free of defects which impair serviceability or detract from its appearance. The Petri dish shall be free of cracks, breaks, chips, sharp edges that tend to cut the hand, and deformation. The plastic material shall be optically clear and sufficiently scratch-free so as not to interfere with the studies of the culture. Mold marks and manufacturer's identification markings on the Petri dish shall be located in positions that will not interfere with studies of the culture. The sealed bag containing the Petri dishes (see 5.1.2.1) shall be free of cuts, tears and pinholes.

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NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETERS (MM).
2. DIAMETER OF CIRCULAR SURFACE TO PROVIDE CLEARANCE WITH OUTSIDE RIM OF DISH.
3. DO NOT SCALE DIMENSIONS.

FIGURE 1. Dish, Culture, Petri.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I. The unit of product for inspection purposes shall be the individual Petri dish or bag (container) of dishes as indicated in 4.3.1 and 4.3.3. Not more than 10 individual Petri dishes shall be selected from a unit container. All samples shall be randomly selected.

TABLE I. For sampling.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-2	2.5
For count	S-2	2.5

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4.2.2 For tests. Sampling for tests shall be conducted in accordance with requirements of 4.4.1 for compliance with 3.3. The acceptance number shall be zero.

4.3 Examination. The Petri dishes shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with table II. Examination shall not be restricted to the possible defects listed in the table.

TABLE II. Classification of defects.

Categories	Defects
A. Unit of product - Petri dish	
Major	
101	Petri dish not free of cracks.
102	Petri dish not free of breaks.
103	Petri dish not free of chips.
104	Petri dish not free of sharp edges that tend to cut the hand.
105	Markings not easily applied on exterior surfaces.
106	Petri dish not clean or dry.
107	Petri dish not properly assembled.
108	Petri dish not free of deformation.
109	Petri dish not free of crazing, which impairs visibility.
110	Petri dish not transparent or not clear within functional requirement of usage.
111	Removal or replacement of top section not easily accomplished with one hand.
Minor	
201	Identification marking not legible, complete or correct.
202	Petri dish not free of burrs or flashes.
203	Petri dish not free of pits, scratches or abrasions which impair visibility in normal usage.
204	Petri dish not free of embedded foreign material.
205	Petri dish not free of loose plastic particles or other foreign material.
B. Unit of product - Bag of dishes (see 5.1.2.1)	
Major	
301	Bag not properly sealed to maintain sterility of contents.
302	Bag not free of cuts, tears, pinholes or openings.
Minor	
401	Required marking (see 5.3.1) not present or legible.

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4.3.2 Dimensional examination. The Petri dish shall be examined for defects in dimensions. Any dimension not within specified limits shall be classified as a defect.

4.3.3 Count. The contents of a bag or container (see 5.1.2.1 or 5.1.3 as applicable) shall be checked for a sufficient quantity of Petri dishes. Any container (bag) not containing the minimum number of Petri dishes as offered in accordance with 5.1.2.1 or 5.1.3, shall be classified as a defect.

4.3.4 Packaging inspection. The inspection of the unit packaging, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Sterility test. Petri dishes shall be tested for sterility in accordance with the method specified in the USP Procedural Details for Sterility Tests for Non-Parenteral Devices, using inoculated product or paper spore strip (carrier). Product containing a biological indicator shall be marked or labeled so that it clearly differentiates from product intended for distribution. The incubation medium and minimum time in each case shall be soybean casein digest for 7 days from inoculated product or paper spore strip (carrier). The incubation temperature shall be as indicated in 4.4.1.1. The minimum number of test units, regardless of lot size, shall be 10, except for inoculated carrier which shall be 20 for Type I Sterilization and 30 for Type II Sterilization, including inoculated carrier and product test units. The inspection level shall be S-3 in accordance with MIL-STD-105. Unless othewise specified, the product for testing purposes shall be that quantity of Petri dishes that will be supplied in the unit package (see 5.1.1).

4.4.1.1 Inoculated product. For each type of sterilization, the organism used shall be as listed below:

Type of sterilization	Organism	Incubation temperature
Steam	B. stearothermophilus	55° - 65° C
Dry heat	B. stearothermophilus, or	55° - 65° C
	B. subtilis var. niger (or globigii)	30° - 35° C
Gas	B. subtilis var. niger (or globigii)	30° - 35° C
Radiation	B. pumilus	30° - 35° C

The number of viable cells shall be sufficient to provide assurance that the product has been sterilized.

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4.4.1.2 Inoculated carrier. Inoculated carriers (paper spore strips) if used shall be inoculated to be of equivalent resistance to spore populations included in MIL-S-36586. The method of incubation and number of test units shall be as indicated in 4.4.1 and 4.4.1.1. The method of sterilization will control the organism in the spore strip to be tested for as shown in 4.4.1.1. NOTE: When an inoculated carrier is chosen, product samples must also be tested as required by the USP.

4.4.1.3 Alternate method. Suppliers conducting sterility tests or designating lots in a manner other than that indicated herein shall submit proof that the alternate method provides equal assurance concerning the sterility of the product. However, all sterility tests shall be conducted using biological indicators. Requests for approval of any alternate method of testing for sterility shall be made to the contracting officer, and approval obtained before employment of the alternate method on a Government contract.

4.4.1.4 Product testing. Testing for sterility of the product after receipt by the Government shall be without a biological indicator in accordance with the USP.

5. PACKAGING.

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.1.1 Unit of issue. One package containing one hundred disposable Petri dishes, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit packaging. Petri dishes shall be clean, dry and packaged sterile in a clean, heat sealed, plastic bag of an appropriate size and design and being capable of maintaining sterility of contents unless the bag is damaged or opened. Ten plastic bags containing ten dishes per bag, or five plastic bags containing twenty dishes per bag, or four plastic bags containing twenty-five dishes per bag, shall be further packaged in a box conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.1.3 Level C. Petri dishes shall be clean, dry and packaged sterile in containers capable of maintaining sterility of contents unless damaged or opened. Ten containers containing ten dishes per container, or five containers containing twenty dishes per container, or four containers containing twenty-five dishes per container shall be further packaged in a manner that will afford adequate protection against deterioration or physical damage during shipment from the supply source to the first receiving activity.

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5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. Four unit packages shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Four unit packages shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged dishes shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carriers rules and regulations.

5.3 Marking.

5.3.1 Plastic bags. In addition to any commercial markings, each bag shall be marked with the lot or control number and the legend or similar statement: "STERILITY GUARANTEED UNLESS BAG IS DAMAGED OR OPENED". Marking may be printed on the bag or on a label inserted in the bag in such a manner so as to be readily seen.

5.3.2 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. Lot (control) number, contract or purchase order number and name and address of the contractor shall be shown. The method of sterilization (G-Gas, R-Radiation, C-Cold, S-Steam, D-Dry heat sterilization) shall be shown following the level and date of packaging. Example: "A-3/77-G." Final sterilization shall be on or after indiated packaging date.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Lot (control) number and the method of sterilization as specified in 5.3.2 shall be shown.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number and date of this specification.
b. National Stock Number required.
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.2 National Stock Number (NSN) Coverage. This specification covers the following item appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6640-01-031-3139	DISH, CULTURE, PETRI, Top and Bottom, Complete, Disposable, 100s.

6.3 Limitations. This specification does not cover all types or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover those items which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DoD - MB

Preparing activity:
DLA-DM

Project No. 6640-1013

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-D-37738 DISH, CULTURE, PETRI, Top and Bottom, Complete, Sterile, Disposable

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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